

PRATHEEN REDDY PININITI

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AI Engineer & Research Analyst | GenAI • ML • Agentic Systems • Edge AI

PROFESSIONAL SUMMARY

M.S. Artificial Intelligence candidate (GPA 3.9, FAU 2026) with hands-on experience building GenAI proof-of-concepts, agentic AI systems, and ML pipelines. Proven ability to translate complex business workflows into AI-enabled solutions using LLMs, RAG, prompt engineering, and cloud platforms (Azure/AWS). Published SPAD camera researcher with Canon collaboration. Passionate about responsible AI adoption, stakeholder enablement, and driving measurable business value through emerging technologies.

EDUCATION

M.S. Artificial Intelligence | GPA: 3.9 Aug 2024 – May 2026
Florida Atlantic University — Boca Raton, FL
Thesis: SPAD Camera Image Analysis — DN-Domain Noise Characterization of the Canon MS-500 (Order No. 32701854)

B.Tech Computer Science (AI & ML) | GPA: 3.7 May 2024
Lovely Professional University — India

EXPERIENCE

Graduate Research Assistant — AI & Imaging Systems Aug 2024 – May 2026
Florida Atlantic University × Canon Inc. (Collaboration) — Boca Raton, FL

- Developed a 5-tier Python ML pipeline (dark noise, PRNU, inter-pixel correlation, PSD, PCA/ICA) for Canon MS-500 SPAD sensor; reduced characterization time by 60% vs. manual methods.
- Built proof-of-concept agentic data analysis tools using LangChain + ReAct to automate iterative sensor diagnostics, simulating AI-enabled business process automation.
- Designed reusable reference architectures and analysis playbooks adopted by research team; results published in formal thesis for Canon stakeholder review.
- Collaborated cross-functionally with Canon engineers (Kevin Fogarty) and faculty committee to align AI-enabled deliverables with business requirements and Responsible AI principles.

Teaching Assistant — Machine Learning & Communication Networks Aug 2024 – May 2026
Florida Atlantic University — Boca Raton, FL

- Facilitated ML labs (regression, classification, deep learning) for 60+ students; created enablement assets and reusable curriculum materials to accelerate AI adoption.
- Graded and provided analytical feedback on advanced ML coursework, reinforcing best practices in model evaluation, data analytics, and responsible model deployment.

Founder & AI Engineer — Auraltr (Fashion Tech Startup) May 2026– Present
Self-Directed / Startup — Miami, FL

- Engineered a GenAI virtual try-on MVP (React Native, FASHN API, CLIP embeddings, Supabase) integrating computer vision and LLM-based recommendations for personalized fashion retail.
- Designed end-to-end AI-enabled product workflow: user onboarding → CLIP semantic search → GenAI try-on → affiliate monetization (AWIN); built POC to production in 8 weeks.
- Implemented Azure-compatible Supabase backend with role-based data governance and secure API key management aligned with responsible AI and data security frameworks.

TECHNICAL SKILLS

AI / GenAI	LLMs (LLaMA, Mistral), GenAI POCs, Prompt Engineering, RAG, Agentic AI (LangChain, LangGraph, ReAct), LoRA/QLoRA Fine-Tuning	CLIP, SAM-2, MediaPipe, Real-ESRGAN, NAFNet, YOLOv8, Computer Vision
Cloud & Data	Azure, AWS (Bedrock, SageMaker), BigQuery, Dataflow/Beam, Power BI, Supabase, Firebase	ETL Pipelines, Data Visualization, pgvector, SQL, Python (NumPy, Pandas, SciPy, Matplotlib)
Dev & Frameworks	Python, React Native, FastAPI, TypeScript, Flutter, Three.js, PyTorch, TensorFlow	Stakeholder Collaboration, Change Management, Responsible AI, Agile, Project Management

KEY PROJECTS & POCS

Warroom — Multi-Agent AI Platform (ReAct + RAG + LangGraph)	2025
Built a multi-agent orchestration platform with ReAct reasoning, RAG retrieval, and LangGraph state management; designed reusable agent playbooks for business automation use cases.	
NeuroPulse — Stroke Risk Prediction App (XGBoost + Three.js)	2025
Deployed ML-powered clinical risk model with 3D brain visualization (Blender + Three.js) and explainability layer; aligned output with responsible AI transparency principles.	
FocusFeed — AI-Powered Android Focus App (Firebase Remote Config)	2025
Developed Android productivity app using AI-driven overlay system, Firebase remote config, and Room database; demonstrated ability to ship AI-enabled MVPs on accelerated timelines.	

PUBLICATIONS & CERTIFICATIONS

Thesis Published: SPAD Camera Image Analysis — Canon MS-500 (Order No. 32701854)	2026
Google Advanced Data Analytics Professional Certificate	2025
Google UX Design Professional Certificate	2025